Identifiability of Deep Normalized Attention: Or, Machine Verified Proofs for Neuroalgebraic Geometry

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Abstract

Identifiability of deep normalized attention is not ensured by positivity and injectivity of the scalar normalizer: we construct a positive injective Borel normalizer for which every point of an explicit open dense set of scalar depth-two parameters has a distinct partner with exactly the same function realization. For exponential softmax, however, we prove that the virtual weights are uniquely determined at every finite depth and every token count $t \geq 2$, provided that the final linear map is nonzero and the score matrices before the final attention layer are not skew-symmetric. The arbitrary-depth proof, developed with Codex and formally verified in Lean 4, uses inputs with one distinguished token and repeated copies of another. At the first layer where two networks have different score matrices, suitable probes produce a nonzero leading term in the displacement between corresponding output tokens as the distance between the incoming token classes tends to zero; the outgoing distance within each network is subdominant, so later normalized layers cannot cancel the leading displacement.

Keywords: parameter identifiability, functional fibers, normalized self-attention

1. Introduction

Henry, Marchetti, and Kohn conjectured that a deep normalized self-attention network is identifiable from its realized function after passing to the virtual parametrization they introduced (Henry et al., 2025). Their lightning, or unnormalized, model retains layerwise rescaling symmetry even after primitive query-key factorizations and changes of hidden-state coordinates have been absorbed into the virtual weights; restoring row normalization suggested that the remaining rescalings should disappear. The resulting conjecture, and the numerical evidence that softmax breaks the lightning rescaling symmetry, were later highlighted in the broader review of neural-network parameter symmetries by Zhao et al. (2025). In the same single-head attention setting, write W for the virtual weights and let the realization map send W to the function ϕ_W^S that it defines on token matrices, where $S : \mathbb{R} \to \mathbb{R}_{>0}$ is injective and $S(0) = 1$ (Henry et al., 2025). The complete functional fiber of W is the set of all virtual parameters that identically realize ϕ_W^S on the whole input space. Henry et al. (2025) proved that this fiber is a singleton at one layer for tokens $t \geq 2$ when the value map is nonzero, conjectured generic singleton fibers at arbitrary depth, and supported the conjecture by a sampled-Jacobian calculation at depth two. We resolve that

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conjecture in the virtual parametrization in which it was posed: compositions of single-head attention layers without residual connections, layer normalization, feed-forward modules, biases, or positional encodings.

Composing normalized attention layers makes the deep problem different from the one-layer theorem: each row retains only ratios of scalar weights, and a later score matrix is evaluated only on token configurations produced by the earlier normalized layers. We construct a positive injective Borel normalizer with $S(0) = 1$ and $S(4z) = S(z)/4$ for $z \neq 0$.

Consider a scalar two-layer, two-token network with score parameters m_1, m_2 and final linear map l, where $m_1 m_2 l \neq 0$. The construction ensures that the first layer sends every nonzero input to two nonzero tokens, so $m_2 \neq 0$ makes every second-layer score nonzero. Replacing m_2 by $4m_2$ therefore multiplies every unnormalized weight in each second-layer row by $1/4$, which cancels under normalization. The two networks also agree at zero, so the distinct tuples (m_1, m_2, l) and $(m_1, 4m_2, l)$ realize the same function. Every point with $m_1 m_2 l \neq 0$ thus lies in a non-singleton fiber, and these points form the open dense complement of the three coordinate hyperplanes.

For standard fixed-temperature softmax, the preceding nonidentifiability cannot occur under the following explicit conditions: the final linear map is nonzero and each score matrix before the final attention layer is not skew-symmetric. At every finite depth and every token count $t \geq 2$, the realized function then uniquely determines all virtual weights. This uniqueness excludes every other parameter point of the same dimensions and depth, without assuming that the other point satisfies the same nondegeneracy conditions, and the hypotheses hold on an explicit open dense set. The proof first determines the final linear map from constant inputs, then compares two networks on inputs formed from one distinguished token and repeated copies of another. At the first differing score matrix, the two networks receive the same two token classes. As the distance s between these classes tends to zero, suitable probes produce a nonzero leading s^2 term in the displacement between corresponding output tokens, while the outgoing distance within each network is $O(s^3)$. Subsequent normalized layers average tokens within these smaller gaps and therefore preserve the leading displacement. The probes are chosen so that the final linear map does not annihilate the leading displacement, allowing the score matrices to be identified successively through the depth.

2. Related work

Exact recovery from a realized function has been established for specific classes of feedforward networks. Minimal scalar-output networks with one hidden tanh layer are identifiable up to their finite structural symmetries (Sussmann, 1992), while analytic continuation recovers generic fully connected deep tanh networks (Fefferman, 1994). Vlačić and Bölcskei (2022, Theorem 1) construct a family of activations for which networks of arbitrary depth and connectivity are identifiable up to hidden-neuron permutations under their nondegeneracy and no-clones hypotheses. For ReLU networks, Phuong and Lampert (2020) study functional equivalence modulo positive rescalings and neuron permutations under architectural and geometric restrictions, while Stock and Gribonval (2023) construct an embedding invariant under positive neuron rescaling.

Neuron permutations and positive rescalings need not exhaust the functional fibers of ReLU networks (Grigsby et al., 2023). At the infinitesimal level, functional dimension measures variation of the realized function through ranks of finite-batch evaluation Jacobians (Grigsby et al., 2025; Grigsby and Lindsey, 2024). For depth-two softmax networks, the sampled Jacobian ranks reported by Henry et al. (2025) provide local evidence for the normalized attention conjecture, but even injectivity of the realization map in a neighborhood of a virtual parameter point does not exclude another point outside that neighborhood realizing the same function on every input.

Brunner et al. (2020) show that distinct attention weights can produce the same head output for a fixed input and fixed value vectors. For scalar-output attention regressors, Bhattamishra et al. (2026) recover a single head's score matrix and nonzero value vector from exact input-output queries and exhibit multihead nonuniqueness when heads share a score matrix; Kim and Kim (2026) recover canonical multihead score-matrix/value-vector pairs up to head permutation using exact variable-length queries. For primitive multihead parameters, Tran et al. (2025) characterize functional equivalence under nondegeneracy conditions over inputs of every sequence length, while Henry (2026) classifies generic fibers at a fixed token count $t \geq 3$. Tran et al. (2026) show that additive sinusoidal encodings preserve functional equivalence, whereas rotary encodings restrict the query-key symmetry group.

For a single lightning-attention layer, Alexandr et al. (2026) study polynomial relations among the coefficients of the realized function. At arbitrary depth, Henry, Marchetti, and Kohn (2025) pass to virtual weights, removing primitive factorization and hidden-coordinate ambiguities. Their lightning fibers retain layerwise rescalings, and their normalized-attention conjecture asks whether the entire input-output function generically determines the virtual weights uniquely. We resolve this conjecture, constructing non-singleton fibers for an admissible normalizer and proving generic singleton fibers for fixed-temperature softmax.

3. Virtual normalized attention and functional fibers

Let $d, t \geq 1$ be integers denoting the embedding dimension and token count, respectively. Write $X = (x_1, \ldots, x_t) \in \mathbb{R}^{d \times t}$ for an indexed sequence of token columns $x_i \in \mathbb{R}^d$. For a score matrix $M \in \mathbb{R}^{d \times d}$ and a positive normalizer $S : \mathbb{R} \to \mathbb{R}_{>0}$, define the layer action through its ith output token column,

$$[\mathsf{T}^S_M(X)]_i = \frac{\sum_{j=1}^t S(x_i^\top M x_j) x_j}{\sum_{j=1}^t S(x_i^\top M x_j)}, \tag{1}$$

where each position $i = 1, \ldots, t$ attends to every input token, including $j > i$, so the model has no temporal or causal masking. For positive integers ℓ and d_{out}, we consider depth ℓ and output dimension d_{out} normalized attention networks realized by the composition

$$\phi^S_W(X) = \big(L \circ \mathsf{T}^S_{M_\ell} \circ \cdots \circ \mathsf{T}^S_{M_1}\big)(X), \tag{2}$$

where

$$W = (M_1, \ldots, M_\ell, L) \in (\mathbb{R}^{d \times d})^\ell \times \mathbb{R}^{d_{\text{out}} \times d} = \mathcal{P}_{d, d_{\text{out}}, \ell} \tag{3}$$

is the tuple of virtual weights. The final map $L : \mathbb{R}^d \to \mathbb{R}^{d_{\text{out}}}$ acts on each token. For a fixed normalizer S and token count t, we study the complete virtual fiber of W,

$$\mathcal{F}_S(W) = \{W' \in \mathcal{P}_{d,d_{\text{out}},\ell} : \phi^S_{W'} = \phi^S_W\}, \tag{4}$$

consisting of all virtual parameter tuples of the same dimensions and depth that realize the same function on every input token matrix; identifiability is precisely the condition $\mathcal{F}_S(W) = \{W\}$, under which equality of the realized functions forces equality of every score matrix and the final linear map. In the notation of Henry et al. (2025, Section 3.3, Eq. (8)), let V_r be the value map and A_r the primitive score matrix at layer r, and set $L_r = V_r \cdots V_1$. Then $M_1 = A_1$, $M_r = L_{r-1}^\top A_r L_{r-1}$ for $r > 1$, and $L = L_\ell$. Moving these cumulative value maps through the normalized layers gives Equation (2), so Conjecture 3.10 concerns the virtual fiber after the primitive factorization data have been absorbed.

For a fixed query token $x_i \in \mathbb{R}^d$, let $s_{ij} = x_i^\top M x_j$ denote the score assigned to key x_j, and let p_{ij} denote the corresponding normalized attention weight. Then

$$\frac{p_{ij}}{p_{ik}} = \frac{S(s_{ij})}{S(s_{ik})}, \tag{5}$$

so normalization retains the ratios among $S(s_{i1}), \ldots, S(s_{it})$, but discards their common scale: multiplying all these values by the same positive constant leaves the attention row unchanged. Injectivity of S permits recovery of s_{ij} from $S(s_{ij})$, but does not by itself permit recovery of the scores from a normalized attention row. In contrast, exponential softmax $S_\tau(z) = e^{z/\tau}$, with fixed $\tau > 0$, implies $\tau \log(p_{ij}/p_{ik}) = s_{ij} - s_{ik}$, so the normalized weights determine the score differences within each row. For one layer, $t \geq 2$, and a nonzero value map, the realized function determines the score matrix (Henry et al., 2025). At greater depth, however, each subsequent score matrix is evaluated on token configurations produced by the preceding normalized layers; identifiability thus depends on whether, under the successive action of the normalized layers, distinct sequences of score matrices produce distinct input-output functions.

Theorem 1 (Counterexample to Conjecture 3.10 of Henry et al. (2025)) *There exists a positive injective Borel map $S : \mathbb{R} \to \mathbb{R}_{>0}$ with $S(0) = 1$ such that, for $d = d_{\text{out}} = 1$, $t = 2$, and $\ell = 2$, every $W = (m_1, m_2, l)$ with $m_1 m_2 l \neq 0$ has the distinct partner $W^\sharp = (m_1, 4m_2, l)$ obeying $\phi^S_{W^\sharp} = \phi^S_W$. Moreover, the parameter set $\{m_1 m_2 l \neq 0\}$ is nonempty, Euclidean open, and dense.*

Proof Row normalization is unchanged when every unnormalized weight in a row is multiplied by the same positive constant. For $x \neq 0$, let $q(x) = \log_4 |x|$, $k(x) = \lfloor q(x) \rfloor$, and $r(x) = q(x) - k(x) \in [0, 1)$. Define

$$S(x) = 4^{g(x)}, \text{ where } \quad g(x) = \begin{cases} -k(x) + \frac{3}{4} + \frac{1}{8} r(x), & x > 0, \\ -k(x) - \frac{3}{4} + \frac{1}{8} r(x), & x < 0, \\ 0, & x = 0, \end{cases} \tag{6}$$

with $k(x)$ specifying the magnitude interval $4^{k(x)} \leq |x| < 4^{k(x)+1}$ and $r(x)$ determining the relative magnitude through $|x|/4^{k(x)} = 4^{r(x)}$. The fractional part of $g(x)$ lies in $[3/4, 7/8)$ for

$x > 0$ and $[1/4, 3/8)$ for $x < 0$, thereby determining the sign of x. Appendix A shows that x can be recovered from $g(x)$, proving injectivity of g and hence of S. For $x \neq 0$, multiplication by four increases $k(x)$ by one while preserving $r(x)$ and the sign, so $g(4x) = g(x) - 1$ and $S(4x) = S(x)/4$. Positivity and $S(0) = 1$ follow from the definition; Appendix A verifies Borel measurability.

Let $(a, b) \neq (0, 0)$ be an input and $(y_1, y_2) = \mathsf{T}^S_{m_1}(a, b)$ its first-layer output. The second-layer scores are $m_2 y_i y_j$, $i, j \in \{1, 2\}$, so $m_2 \neq 0$ reduces their nonvanishing to $y_1 y_2 \neq 0$. Appendix A proves $zS(z) > wS(-w)$ for all $z, w > 0$, which excludes cancellation in either first-layer row when $ab < 0$ and $m_1 \neq 0$. If $ab \geq 0$, the tokens have the same weak sign and at least one is nonzero, so every positive weighted average of them is nonzero. Thus $y_1 y_2 \neq 0$.

Since W and $W^\sharp$ differ only in their second score matrix, they have the same first-layer output (y_1, y_2). For every nonzero input, $m_2 y_i y_j \neq 0$ for all i, j, and hence $S(4m_2 y_i y_j) = S(m_2 y_i y_j)/4$. Replacing m_2 by $4m_2$ therefore multiplies every unnormalized weight in each second-layer row by the same factor $1/4$, which cancels under row normalization. The zero input agrees separately, and the common final map l preserves equality of the two realizations. Because $m_2 \neq 0$, the parameter points are distinct. Finally, $\{m_1 m_2 l \neq 0\}$ is the complement of the coordinate hyperplanes $m_1 = 0$, $m_2 = 0$, and $l = 0$, and is therefore nonempty, open, and dense. ∎

Remark 1. For a one-layer scalar network with the normalizer of Theorem 1, score parameter $m \neq 0$, and final map $l \neq 0$, an input $(a, 0)$ with $a \neq 0$ gives the query a one nonzero score ma^2 and one zero score. Replacing m by $4m$ scales $S(ma^2)$ by $1/4$ while leaving $S(0) = 1$ fixed, so the normalized weight on a, and hence the output, changes, consistently with the one-layer identifiability theorem of Henry et al. (2025). In the depth-two construction, the first layer sends every nonzero input to two nonzero tokens, so replacing m_2 by $4m_2$ scales every unnormalized second-layer weight by $1/4$ and preserves the normalized weights.

Remark 2. The normalizer in Theorem 1 satisfies all of the assumptions on S in Conjecture 3.10 of Henry et al. (2025, Section 3.4): it is positive, injective, and satisfies $S(0) = 1$. It is also Borel measurable. The normalizer is discontinuous at zero and unbounded near zero, so the counterexample does not establish nonidentifiability for smooth normalizers; we leave open whether continuity together with the hypotheses of Henry et al. (2025) is sufficient for generic identifiability.

Theorem 2 (Identifiability for exponential softmax) *Let $d, d_{\text{out}} \geq 1$, $t \geq 2$, $\ell \geq 1$, and $\tau > 0$, and set $S_\tau(z) = e^{z/\tau}$. If $W = (M_1, \ldots, M_\ell, L) \in \mathcal{P}_{d,d_{\text{out}},\ell}$ satisfies a nonvanishing condition on its final map and non-skewness conditions on the score matrices before its final attention layer,*

$$L \neq 0 \quad \text{and} \quad M_r^\top \neq -M_r \quad \text{for } 1 \leq r < \ell, \tag{7}$$

then

$$\phi^{S_\tau}_{W'} = \phi^{S_\tau}_{W} \quad \Longrightarrow \quad W' = W, \tag{8}$$

for every $W' \in \mathcal{P}_{d,d_{\mathrm{out}},\ell}$, without a nondegeneracy assumption on W'. Moreover, for any $1 \le i \le d$ and $1 \le o \le d_{\mathrm{out}}$, the set

$$\mathcal{U}_{oi} = \left\{ W \in \mathcal{P}_{d,d_{\mathrm{out}},\ell} : L_{oi} \prod_{r=1}^{\ell-1} (M_r)_{ii} \neq 0 \right\}, \tag{9}$$

is nonempty, Euclidean open and dense, is contained in the set specified by Equation (7), and satisfies $\mathcal{F}_{S_\tau}(W) = \{W\}$ for every $W \in \mathcal{U}_{oi}$; here the product is understood to equal one when $\ell = 1$.

Proof Without loss of generality, let $\tau = 1$. Suppose $\phi_W^{S_\tau} = \phi_{W'}^{S_\tau}$, with $W' = (M_1', \dots, M_\ell', L')$ otherwise arbitrary. We call the network with virtual parameters W the source network and the network with virtual parameters W' the competing network.

Our identification of the virtual weights has five steps, with M_r and M_r' compared successively for $r = 1, \dots, \ell$, namely, 1) constant token sequences determine the final linear map because every attention layer fixes identical tokens and the network output repeats their image under that map; 2) once $M_j = M_j'$ for every $j < r$, Lemma 1 keeps the two token classes of $(x + \varepsilon u, x - \varepsilon u, \dots, x - \varepsilon u)$, $\varepsilon \neq 0$ and $u \neq 0$, distinct through layers $1, \dots, r-1$ whenever $u^\top M_j u \neq 0$ for every $j < r$; 3) at layer r, Lemma 2 computes the quadratic coefficient of the displacement between corresponding output tokens of the two networks and bounds each network's internal token gap at cubic order in the incoming gap; 4) Lemma 3 preserves this quadratic coefficient through arbitrary, possibly different score matrices at layers $r+1, \dots, \ell$ because each normalized layer takes convex combinations within its input token segment; 5) Lemma 4 supplies a dense set of directions u satisfying $Lu \neq 0$ and $u^\top M_j u \neq 0$ for every $j < r$, so equality of the realized functions determines M_r successively for $r = 1, \dots, \ell$;

Since constant token sequences pass unchanged through normalized attention layers

$$\phi_W^{S_\tau}(x, \dots, x) = (Lx, \dots, Lx) \quad \text{and} \quad \phi_{W'}^{S_\tau}(x, \dots, x) = (L'x, \dots, L'x). \tag{10}$$

Thus, evaluating on repeated copies of each standard basis vector e_j gives $Le_j = L'e_j$ and establishes $L = L'$ by identifying every column.

Fix $1 \le r \le \ell$ and suppose inductively that $M_j = M_j'$ for every $j < r$. To compare M_r and M_r', take an arbitrary base point $x \in \mathbb{R}^d$ and choose, using Lemma 4, any direction $u \in \mathbb{R}^d$ satisfying $Lu \neq 0$ and $u^\top M_j u \neq 0$ for every $j < r$. For a nonzero perturbation ε, consider

$$X_\varepsilon(x, u) = (x + \varepsilon u, x - \varepsilon u, \dots, x - \varepsilon u), \tag{11}$$

with one token in the distinguished class and $t - 1$ tokens in the repeated class. The condition $Lu \neq 0$ prevents annihilation of nonzero displacements along u, while Lemma 1 uses $u^\top M_j u \neq 0$ for $j < r$ to keep the two token classes distinct through layers $1, \dots, r-1$.

For token values $x + pu$ and $x + qu$, Lemma 1 gives the layer action

$$\mathsf{T}_M^{S_\tau}(x + pu, x + qu, \dots, x + qu) = (x + p^+ u, x + q^+ u, \dots, x + q^+ u), \tag{12}$$

with the base point, direction, and multiplicities unchanged, and its noncollapse conclusion applies to each M_j with $j < r$ because $u^\top M_j u \neq 0$. The common layers $1, \dots, r-1$ therefore

produce the same scalar coefficients p_ε and q_ε in both networks immediately before layer r, with p_ε belonging to the distinguished class and q_ε to the repeated class. Convexity keeps both coefficients in $[-|\varepsilon|, |\varepsilon|]$, while prefix noncollapse makes their difference $s_\varepsilon = p_\varepsilon - q_\varepsilon$ a nonzero incoming gap that tends to zero with ε.

In the source network, the layer-r tokens have the form $x + y_i u$, while those in the competing network have the form $x + y'_i u$, with $i = 1$ labeling the distinguished class and $i = 2$ the repeated class; both coefficient pairs depend on ε through p_ε and q_ε. For each $i \in \{1, 2\}$, the quantitative estimates following Lemma 2 express the difference between corresponding scalar coefficients as

$$y_i - y'_i = \kappa_n x^\top (M_r - M'_r) u \, s_\varepsilon^2 + O(|\varepsilon| s_\varepsilon^2), \tag{13}$$

where $\kappa_n = n/(1+n)^2$ is the positive multiplicity factor in the quadratic term and $n = t-1$ is the repeated-class multiplicity. Both classes have the same quadratic coefficient, although their $O(|\varepsilon| s_\varepsilon^2)$ remainders can differ. The difference between the distinguished and repeated coefficients in the source network satisfies

$$y_1 - y_2 = \kappa_n (u^\top M_r u) s_\varepsilon^3 + O(|s_\varepsilon|^4); \tag{14}$$

likewise, the internal gap in the competing network satisfies

$$y'_1 - y'_2 = \kappa_n (u^\top M'_r u) s_\varepsilon^3 + O(|s_\varepsilon|^4), \tag{15}$$

so both internal gaps are $O(|s_\varepsilon|^3)$ and subdominant to the quadratic scale in Equation (13).

In each network, every later token remains in the segment spanned by its two layer-r token values. The $O(|s_\varepsilon|^3)$ bound on each segment's diameter therefore also bounds the change in displacement between corresponding tokens through arbitrary remaining layers. Since p_ε and q_ε lie in $[-|\varepsilon|, |\varepsilon|]$, their gap satisfies $|s_\varepsilon| \le 2|\varepsilon|$, which absorbs this contribution into the $O(|\varepsilon| s_\varepsilon^2)$ remainder in the quadratic expansion. For the distinguished first output token, denoted by the subscript 1 and taking values in $\mathbb{R}^{d_{\text{out}}}$, applying the common final map therefore yields

$$[\phi_W^{S_\tau}(X_\varepsilon)]_1 - [\phi_{W'}^{S_\tau}(X_\varepsilon)]_1 = s_\varepsilon^2 \left[\kappa_n x^\top (M_r - M'_r) u + O(|\varepsilon|) \right] Lu, \tag{16}$$

whose left-hand side vanishes by equality of the realized functions. Since $s_\varepsilon \neq 0$ and $Lu \neq 0$, a nonzero value of the ε-independent coefficient $\kappa_n x^\top (M_r - M'_r) u$ would dominate the $O(|\varepsilon|)$ remainder as $\varepsilon \to 0$ and contradict functional equality; positivity of κ_n therefore yields

$$x^\top (M_r - M'_r) u = 0, \tag{17}$$

for every $x \in \mathbb{R}^d$ and every u satisfying $Lu \neq 0$ and $u^\top M_j u \neq 0$ for every $1 \le j < r$. For the polynomial

$$P_r(u) = \|Lu\|^2 \prod_{j=1}^{r-1} u^\top M_j u, \tag{18}$$

the set $\{u \in \mathbb{R}^d : P_r(u) \neq 0\}$ consists exactly of the directions satisfying $Lu \neq 0$ and $u^\top M_j u \neq 0$ for every $1 \le j < r$. Each factor of P_r is not identically zero because $L \neq 0$ and

$M_j^\top \neq -M_j$ for every $1 \leq j < r$. By Lemma 4, $\{u : P_r(u) \neq 0\}$ is dense, so for each fixed x, continuity in u extends the bilinear identity to every direction. Taking $x = e_i$ and $u = e_j$ then identifies each entry of $M_r - M'_r$, establishing $M_r = M'_r$ for the inductive step. At $r = 1$, the prefix is empty, so the same argument starts the induction without a preceding score condition; applying it through layer ℓ establishes $W = W'$ by recovering every score matrix and the final map. Recovery of M_r uses non-skewness only of M_j for $j < r$, so M_ℓ and all competing score matrices $M'_1, \ldots, M'_\ell$ remain unrestricted.

It remains to prove the explicit open-dense clause. The coordinate product $L_{oi} \prod_{r=1}^{\ell-1} (M_r)_{ii}$ is a polynomial in the entries of W and is not identically zero, since it equals one when every matrix entry of W equals one. Lemma 4 therefore shows that its nonvanishing set $\mathcal{U}_{oi}$ is nonempty, Euclidean open, and dense. If $W \in \mathcal{U}_{oi}$, then $L_{oi} \neq 0$ and $(M_r)_{ii} \neq 0$ for every $r < \ell$, which ensures $L \neq 0$ and excludes skew-symmetry of each preceding score matrix because a real skew-symmetric matrix has zero diagonal. The pointwise identifiability result therefore applies at every $W \in \mathcal{U}_{oi}$, giving singleton complete fibers on this open dense set.

For any fixed primitive architecture whose query, key, and value matrices have positive row and column dimensions, substituting the query, key, and value matrices into the coordinate product in Equation (9) gives a nonzero polynomial: choosing every primitive matrix entry strictly positive makes L_{oi} and every $(M_r)_{ii}$ positive. The resulting polynomial has an open dense nonvanishing set in primitive parameter space, whose image under the primitive-to-virtual map is precisely the intersection of $\mathcal{U}_{oi}$ with the architecture's virtual parameter image. Continuity of the primitive-to-virtual map makes this intersection dense in the virtual parameter image, and openness of $\mathcal{U}_{oi}$ makes it relatively open; hence singleton complete fibers hold on a relatively open dense subset of the virtual parameters realized by that architecture. ■

Remark 3. Although the directions used to recover M_r are selected through the quadratic forms $u^\top M_j u$ for $1 \leq j < r$, recovery uses the bilinear identity $x^\top (M_r - M'_r) u = 0$ with x and u independent. The argument therefore determines the full score matrix, including its skew-symmetric component, rather than only its symmetric part.

Remark 4. Theorem 2 imposes one-sided hypotheses. No nondegeneracy is assumed of W' a priori: constant inputs force $L' = L$, while recovery of M_r requires no nondegeneracy assumption on any unrecovered score matrix of W'. On the source side, non-skewness is required only for $M_1, \ldots, M_{\ell-1}$ because recovery of M_r requires noncollapse only through the prefix $M_1, \ldots, M_{r-1}$; the terminal score M_ℓ belongs to none of these prefixes. Likewise $L \neq 0$ ensures that some displacement survives the final map, without requiring L to have full rank.

The full source set defined by Equation (7) is itself Euclidean open and dense: $L = 0$ and each skew-symmetric matrix space are proper closed linear subspaces of their respective finite-dimensional ambient spaces. $\mathcal{U}_{oi}$ is a more explicit algebraic open-dense subset obtained by choosing one coordinate direction that satisfies every required nonvanishing condition simultaneously.

4. Tool and computational resource disclosure

Following the disclosure and verifiability principles advocated by the Leiden Declaration on Artificial Intelligence and Mathematics (Leiden Declaration Working Group, 2026) and discussed by Tao (2026), we record the role of automated tools in this work. Codex was used with Lean 4 for proof search, theorem proving, proof debugging, inspection and summarization of formal code, and manuscript drafting and revision. The authors report extensive use of GPT-5.6 Sol, primarily at very-high reasoning effort, with aggregate inference on the order of tens of millions of tokens. This is an author-reported estimate, not a measurement of the verification commands.

James Amarel is the formalizer and maintainer. The mathematical claims and exposition remain the responsibility of the authors. Supporting Lean code, including the headline theorems, log-affine corollaries, and boundary examples, is organized in the DeepLean root Lake project.[1] Its Challenge/Solution interface exposes the three declarations described in Appendix C. Local compilation, statement comparison, transitive axiom audits, and independent proof replay have distinct roles; none establishes the correspondence between formal statements and mathematical prose without a separate semantic review. The reproducible commands and their current results are documented in `docs/verification/README.md`.

5. Conclusion

Positivity and injectivity of the scalar normalizer do not suffice for identifiability of deep normalized attention. Indeed, row normalization retains ratios rather than individual weights, and we exhibit an admissible normalizer whose ratio structure produces non-singleton virtual fibers on an open dense set. Exponential softmax behaves differently: at every finite depth, complete virtual fibers are singletons under the explicit conditions of Theorem 2. Injectivity of S permits recovery of a scalar score z from $S(z)$, but does not by itself exclude changes in the score matrices that, for every input, multiply the unnormalized weights within each attention row by a common positive factor and hence preserve the realized function.

For softmax, Theorem 2 replaces the local evidence from finite-batch Jacobian ranks for deep normalized attention with a global statement on the complete functional fiber. Equality of the realized function on the whole input space recovers the virtual score matrices successively, even when the competing parameter point is arbitrary within the same virtual parameter space, and the hypotheses hold on an explicit nonempty Euclidean-open dense set. Thus normalization cannot be understood simply as removing a rescaling symmetry from lightning attention: its ratio structure can either create persistent functional ambiguity or eliminate it under composition. In the softmax case considered here, the latter rigidity separates the remaining raw query-key and value-coordinate redundancies from the virtual identifiability of the deep normalized model.

1. Code: https://github.com/lanl/DeepLean.

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Appendix A. The counterexample normalizer and its fiber

This appendix verifies positivity, injectivity, and Borel measurability of the normalizer in Theorem 1 and proves that (m_1, m_2, l) and $(m_1, 4m_2, l)$ realize the same function whenever $m_1 m_2 l \neq 0$.

A.1. Injectivity, measurability, and discontinuity

Write $\{u\} = u - \lfloor u \rfloor$ for the fractional part of a real number u. For $x > 0$, Equation (6) gives $\{g(x)\} = 3/4 + r(x)/8 \in [3/4, 7/8)$, whereas for $x < 0$ the decomposition

$$g(x) = (-k(x) - 1) + \left(\frac{1}{4} + \frac{1}{8} r(x)\right), \tag{19}$$

places $\{g(x)\}$ in $[1/4, 3/8)$. The disjoint intervals determine the sign of x from $\{g(x)\}$, while their exclusion of zero separates $g(0) = 0$ from every $g(x)$ with $x \neq 0$.

For $x > 0$, the integer part of $g(x)$ recovers $k(x) = -\lfloor g(x) \rfloor$, and its fractional part recovers $r(x) = 8(\{g(x)\} - 3/4)$. For $x < 0$, the integer shift in Equation (19) gives $k(x) = -\lfloor g(x) \rfloor - 1$, while the fractional part yields $r(x) = 8(\{g(x)\} - 1/4)$. For $x \neq 0$, reconstruction of the magnitude follows from

$$|x| = 4^{k(x) + r(x)}, \tag{20}$$

which determines x together with its recovered sign and proves injectivity of g. Since $u \mapsto 4^u$ is strictly increasing, $S = 4^g$ is injective, and positivity and $S(0) = 1$ follow from its definition.

On each half-line $x > 0$ and $x < 0$, the function g is obtained from continuous functions, the Borel floor function, and arithmetic operations. Assigning $g(0) = 0$ on the remaining singleton therefore makes g Borel measurable on $\mathbb{R}$, and continuity of exponentiation makes $S = 4^g$ Borel measurable. The sequence $x_n = 4^{-n}$ tends to zero and satisfies $k(x_n) = -n$ with $r(x_n) = 0$, so $S(x_n) = 4^{n+3/4} \to \infty$ while $S(0) = 1$, proving discontinuity at zero.

Positivity, injectivity, and $S(0) = 1$ are the normalizer hypotheses of Henry et al. (2025, Section 3.4); Borel measurability is an additional property of the construction.

A.2. Scaling and domination

For $x \neq 0$, the identity $q(4x) = q(x) + 1$ gives $k(4x) = k(x) + 1$ and leaves $r(x)$ unchanged. Since multiplication by four preserves the sign of x, the definition of g yields $g(4x) = g(x) - 1$, and exponentiation gives

$$S(4x) = \frac{1}{4} S(x), \tag{21}$$

so scaling all scores in a row by four leaves its normalized weights unchanged whenever every score in that row is nonzero. At $z = 0$, the equation $S(4z) = S(z)/4$ fails because $S(0) = 1$, so using a uniform factor $1/4$ for all second-layer weights requires every second-layer score to be nonzero.

For arbitrary $z, w > 0$, the construction also satisfies

$$zS(z) > wS(-w), \tag{22}$$

because the positive and negative branches of g give separated exponent ranges. Indeed, multiplying $S(z)$ by $z = 4^{k(z)+r(z)}$ yields

$$zS(z) = 4^{3/4+9r(z)/8}, \tag{23}$$

whose exponent is at least $3/4$, whereas multiplying $S(-w)$ by $w = 4^{k(w)+r(w)}$ yields $wS(-w) = 4^{-3/4+9r(w)/8}$, whose exponent is strictly less than $3/8$.

A.3. Nonvanishing of the first-layer outputs

Fix a nonzero score parameter m and a nonzero input (a, b), and write $(y_1, y_2) = \mathsf{T}^S_m(a, b)$. The first output is

$$y_1 = \frac{S(ma^2)a + S(mab)b}{S(ma^2) + S(mab)}, \tag{24}$$

whose denominator is positive because S is positive; exchanging a and b exchanges the two outputs, so nonvanishing of the numerator of y_1 for every nonzero input also proves $y_2 \neq 0$. If $ab \geq 0$, the two terms $aS(ma^2)$ and $bS(mab)$ are both nonnegative or both nonpositive, and at least one is nonzero, so their sum cannot vanish.

For $ab < 0$, simultaneous negation of a and b preserves the scores ma^2 and mab and negates the numerator of y_1, so it suffices to consider $a > 0 > b$. When $m > 0$, applying Equation (22) to $z = ma^2 > 0$ and $w = -mab > 0$ and canceling the positive factor ma gives

$$aS(ma^2) > (-b)S(mab), \tag{25}$$

so the numerator $aS(ma^2) + bS(mab)$ of y_1 is positive. When $m < 0$, the positive score is mab and the negative score is ma^2; applying Equation (22) with $z = mab$ and $w = -ma^2$ and canceling the positive factor $-ma$ gives

$$(-b)S(mab) > aS(ma^2), \tag{26}$$

so the numerator $aS(ma^2) + bS(mab)$ of y_1 is negative.

For every $m \neq 0$ and every nonzero input (a, b), the sign analysis and the exchange of the two input tokens establish

$$y_1 y_2 \neq 0, \tag{27}$$

so every second-layer score $m_2 y_i y_j$ is nonzero for $i, j \in \{1, 2\}$ when $m = m_1$ and $m_1 m_2 \neq 0$.

A.4. The non-singleton fiber

Let $W = (m_1, m_2, l)$ with $m_1 m_2 l \neq 0$, and set $W^\sharp = (m_1, 4m_2, l)$. Since both networks have first-layer score parameter m_1, they produce the same first-layer output (y_1, y_2) on each input (a, b). For every nonzero input, Equation (27) and $m_2 \neq 0$ make each second-layer score $m_2 y_i y_j$ nonzero, so Equation (21) gives

$$S(4m_2 y_i y_j) = \frac{1}{4} S(m_2 y_i y_j) \quad \text{for } i, j \in \{1, 2\}, \tag{28}$$

so each row's unnormalized weights acquire the common factor $1/4$, which cancels under normalization and leaves both second-layer outputs unchanged. At $(a, b) = (0, 0)$, both first-layer tokens and both second-layer tokens equal zero, so both realizations vanish. Applying the common final map l therefore yields

$$\phi^S_{W^\sharp} = \phi^S_W, \tag{29}$$

with $W^\sharp \neq W$ because $m_2 \neq 0$, proving that $\mathcal{F}_S(W)$ is non-singleton.

The set $\{m_1 m_2 l \neq 0\} \subset \mathbb{R}^3$ is the complement of the three coordinate hyperplanes, and is therefore nonempty, Euclidean open, and dense.

Appendix B. Supporting results for Theorem 2

Constant inputs determine the final map through Equation (10); recovery of each score matrix then requires a nonzero perturbation that survives the preceding layers and produces a displacement that the remaining layers cannot cancel. We retain $S_\tau(z) = e^{z/\tau}$ with $\tau = 1$, as fixed in the proof of Theorem 2. The four lemmas below establish noncollapse of the two token classes, the quadratic displacement between networks and cubic gap within each network, persistence through arbitrary remaining score matrices, and the density of directions on which the displacement survives the final map.

For a query with one distinguished key and $n \geq 1$ identical repeated keys, the normalized weight on the distinguished key depends on its score difference z from a repeated key through

$$\sigma_n(z) = \frac{e^z}{e^z + n}, \tag{30}$$

which lies strictly between zero and one because every unnormalized weight is positive. The factor n records the repeated keys' total contribution to the normalizing denominator.

B.1. Two-class closure and noncollapse

Lemma 1 (Two-class closure and noncollapse) *Let $t \geq 2$ and $n = t - 1$, and let $M \in \mathbb{R}^{d\times d}$, $x, u \in \mathbb{R}^d$ with $u \neq 0$, and $p, q \in \mathbb{R}$. The input*

$$X(x, u; p, q) = (x + pu, x + qu, \ldots, x + qu), \tag{31}$$

has one distinguished token and n repeated tokens. Write $s = p - q$ for the signed difference of their scalar coefficients, and set $a = x^\top Mu$ and $b = u^\top Mu$, so the distinguished query's score difference is $s(a + pb)$. Its output coefficient is

$$p^+ = q + \sigma_n\big(s(a + pb)\big)s, \tag{32}$$

where the sigmoid factor is the normalized weight on the distinguished input. Every repeated query instead has output coefficient

$$q^+ = q + \sigma_n\big(s(a + qb)\big)s, \tag{33}$$

because its query token is $x + qu$. Consequently the layer action is

$$\mathsf{T}^{S_\tau}_M X(x, u; p, q) = X(x, u; p^+, q^+), \tag{34}$$

with the base point, direction, and multiplicities preserved, and both output coefficients between p and q. If $s \neq 0$ and $b \neq 0$, then $p^+ \neq q^+$, so the two output token values remain distinct.

For any finite list $M_1, \ldots, M_k$ with $k \geq 0$ and $u^\top M_j u \neq 0$ for every $1 \leq j \leq k$, the centered input $X(x, u; \varepsilon, -\varepsilon)$ therefore retains distinct token values through all k layers whenever $\varepsilon \neq 0$. Both final scalar coefficients p_ε and q_ε lie in $[-|\varepsilon|, |\varepsilon|]$, so they tend to zero as $\varepsilon \to 0$ while their difference remains nonzero for every $\varepsilon \neq 0$.

Proof For query $x + pu$, subtracting the score of a repeated key from the score of the distinguished key gives

$$(x + pu)^\top M(x + pu) - (x + pu)^\top M(x + qu) = s(a + pb), \tag{35}$$

which determines the ratio between their unnormalized weights. Factoring $e^{(x+pu)^\top M(x+qu)}$ from the attention row gives distinguished-key weight $\sigma_n(s(a + pb))$; averaging the two token values with this weight gives Equation (32). For query $x + qu$, the score difference is $s(a + qb)$, so the same normalization gives Equation (33). All repeated queries are identical and hence produce the same output token, establishing Equation (34).

Subtracting the two coefficient updates gives

$$p^+ - q^+ = s\left[\sigma_n\big(s(a + pb)\big) - \sigma_n\big(s(a + qb)\big)\right], \tag{36}$$

so noncollapse follows from distinct sigmoid arguments whenever $s \neq 0$. The arguments differ by

$$s(a + pb) - s(a + qb) = s^2 b, \tag{37}$$

which is nonzero when $s \neq 0$ and $b \neq 0$. For $n > 0$, equality $\sigma_n(z) = \sigma_n(z')$ implies $ne^z = ne^{z'}$ by cross multiplication, and hence $z = z'$. Injectivity of σ_n therefore makes the bracket in Equation (36) nonzero, proving $p^+ \neq q^+$.

For the finite list, let $p_j(\varepsilon)$ and $q_j(\varepsilon)$ denote the distinguished and repeated coefficients after j layers, starting with $p_0(\varepsilon) = \varepsilon$ and $q_0(\varepsilon) = -\varepsilon$. Applying the exact noncollapse result successively with $b = u^\top M_j u$ gives

$$p_j(\varepsilon) - q_j(\varepsilon) \neq 0 \quad \text{for } 0 \leq j \leq k \text{ and } \varepsilon \neq 0, \tag{38}$$

so the two token values remain distinct at every layer. Each coefficient update is a convex combination of the incoming coefficients because $0 < \sigma_n(z) < 1$. The coefficient intervals are therefore nested inside $[-|\varepsilon|, |\varepsilon|]$, proving convergence of both coefficients to zero after every layer, including the empty list $k = 0$. ■

At step r of Theorem 2, equality $M_j = M'_j$ for every $j < r$ gives both networks the same coefficients p_ε and q_ε after applying the common prefix to $X(x, u; \varepsilon, -\varepsilon)$. For $\varepsilon \neq 0$ and a direction satisfying $Lu \neq 0$ and $u^\top M_j u \neq 0$ for every $j < r$, Lemma 1 ensures that their signed difference $s_\varepsilon = p_\varepsilon - q_\varepsilon$ is nonzero and satisfies $|s_\varepsilon| \leq 2|\varepsilon|$, so it tends to zero while the token values remain distinct. The parameter ε specifies the original input perturbation, whereas s_ε measures the gap on which the comparison of M_r and M'_r depends.

B.2. Quadratic signal and cubic gap

Lemma 2 (Quadratic signal and cubic gap) *Let $t \geq 2$ and $n = t-1$, and fix $x, u \in \mathbb{R}^d$ with $u \neq 0$ and $N, N' \in \mathbb{R}^{d\times d}$. Suppose p_ε and q_ε tend to zero, with signed gap $s_\varepsilon = p_\varepsilon - q_\varepsilon \neq 0$ for every sufficiently small nonzero ε. Apply $\mathsf{T}_N^{S_\tau}$ to $X(x, u; p_\varepsilon, q_\varepsilon)$ and denote its distinguished output by $x + y_1 u$ and its repeated output by $x + y_2 u$; use y_1' and y_2' for the corresponding coefficients after applying $\mathsf{T}_{N'}^{S_\tau}$ to the same input. These coefficients depend on ε and are uniquely determined because $u \neq 0$.*

Write $a = x^\top N u$ and $a' = x^\top N' u$ for the two matrices' scores between the base point and direction, and write $b = u^\top N u$ and $b' = u^\top N' u$ for their quadratic scores on the direction. All four output coefficients tend to zero. The positive coefficient $\kappa_n = \sigma_n'(0) = n/(1+n)^2$ determines the displacement for each corresponding token class through

$$y_i - y_i' = \kappa_n (a - a') s_\varepsilon^2 + o(s_\varepsilon^2) \quad \text{for } i \in \{1, 2\}, \tag{39}$$

where both classes have the same quadratic coefficient, although their remainders can differ. Within the output of the layer with matrix N, the class gap satisfies

$$y_1 - y_2 = \kappa_n b\, s_\varepsilon^3 + O(|s_\varepsilon|^4), \tag{40}$$

whereas the layer with matrix N' satisfies

$$y_1' - y_2' = \kappa_n b'\, s_\varepsilon^3 + O(|s_\varepsilon|^4), \tag{41}$$

so both internal gaps are of smaller order than the cross-network quadratic term whenever $a \neq a'$, including when b or b' vanishes.

Proof For $i \in \{1, 2\}$, let $c_1 = p_\varepsilon$ denote the distinguished input coefficient and $c_2 = q_\varepsilon$ the repeated input coefficient. Lemma 1 gives

$$y_i = q_\varepsilon + \sigma_n\big(s_\varepsilon(a + c_i b)\big) s_\varepsilon, \tag{42}$$

because the sigmoid factor is the normalized weight on the distinguished key for that class's query. Replacing (a, b) by (a', b') gives the coefficient y_i' for the layer with matrix N'. Each output coefficient lies between p_ε and q_ε, so all four tend to zero.

The weighted sigmoid has derivative

$$\sigma_n'(z) = \frac{n e^z}{(e^z + n)^2}, \tag{43}$$

whose value at zero is κ_n. Since σ_n'' is bounded on a neighborhood of zero, integrating σ_n' between arguments A and B in that neighborhood gives

$$\sigma_n(A) - \sigma_n(B) = (A - B)\left[\kappa_n + O(|A| + |B|)\right], \tag{44}$$

with the same identity when $A = B$. All arguments in the coefficient updates are $O(|s_\varepsilon|)$ because p_ε and q_ε tend to zero. For corresponding classes, the argument difference is $s_\varepsilon[(a - a') + c_i(b - b')]$, so the outer factor s_ε in Equation (42) gives

$$y_i - y_i' = \kappa_n[(a - a') + c_i(b - b')] s_\varepsilon^2 + O(|s_\varepsilon|^3), \tag{45}$$

whose class-dependent quadratic correction is $o(s_\varepsilon^2)$ since $c_i \to 0$. The cubic remainder is also $o(s_\varepsilon^2)$, proving Equation (39).

For the two classes within the layer with matrix N, the sigmoid arguments instead differ by

$$s_\varepsilon(a + p_\varepsilon b) - s_\varepsilon(a + q_\varepsilon b) = s_\varepsilon^2 b, \tag{46}$$

so Equation (44) gives their sigmoid difference as $s_\varepsilon^2 b[\kappa_n + O(|s_\varepsilon|)]$. Multiplication by the outer s_ε in the exact gap update Equation (36) proves Equation (40), with its signed cubic term and fourth-order remainder. Applying the same argument to N' proves Equation (41); if either quadratic score b or b' is zero, the corresponding exact internal gap is zero as well. ■

For the centered probes in Theorem 2, Lemma 1 places both incoming coefficients in $[-|\varepsilon|, |\varepsilon|]$, so $c_i = O(|\varepsilon|)$ and $|s_\varepsilon| \le 2|\varepsilon|$. The class-dependent correction and cubic remainder in Equation (45) therefore give the sharper comparison

$$y_i - y_i' = \kappa_n x^\top (N - N')u\, s_\varepsilon^2 + O(|\varepsilon| s_\varepsilon^2), \tag{47}$$

whose remainder is dominated by the quadratic term whenever $x^\top(N - N')u \neq 0$. Both internal class gaps remain $O(|s_\varepsilon|^3)$, so their effect on later outputs is smaller than this quadratic displacement.

B.3. Persistence through unequal suffixes

Lemma 3 (Persistence through unequal suffixes) *Let $t \ge 2$ and $x, u \in \mathbb{R}^d$ with $u \neq 0$. Suppose the scalar coefficients $p_\varepsilon, q_\varepsilon, p'_\varepsilon, q'_\varepsilon$ all tend to zero, and let $\delta_\varepsilon \to 0$ be nonzero for every sufficiently small nonzero ε. For a fixed $w \in \mathbb{R}$, suppose the distinguished coefficients satisfy $p_\varepsilon - p'_\varepsilon = w\delta_\varepsilon + o(|\delta_\varepsilon|)$ and the repeated coefficients satisfy $q_\varepsilon - q'_\varepsilon = w\delta_\varepsilon + o(|\delta_\varepsilon|)$. Assume also that $p_\varepsilon - q_\varepsilon = o(|\delta_\varepsilon|)$ within the unprimed network and $p'_\varepsilon - q'_\varepsilon = o(|\delta_\varepsilon|)$ within the primed network.*

Apply layers with arbitrary score matrices $M, M' \in \mathbb{R}^{d\times d}$ to $X(x, u; p_\varepsilon, q_\varepsilon)$ and $X(x, u; p'_\varepsilon, q'_\varepsilon)$, respectively. Denote the distinguished and repeated output coefficients by y_1 and y_2 after $\mathsf{T}_M^{S_\tau}$, and by y'_1 and y'_2 after $\mathsf{T}_{M'}^{S_\tau}$. All four output coefficients tend to zero, and corresponding classes satisfy

$$y_i - y_i' = w\delta_\varepsilon + o(|\delta_\varepsilon|) \quad \text{for } i \in \{1, 2\}, \tag{48}$$

with internal gap $y_1 - y_2 = o(|\delta_\varepsilon|)$ in the unprimed network and $y'_1 - y'_2 = o(|\delta_\varepsilon|)$ in the primed network. The same conclusions hold after any two finite lists of softmax layers of equal length, with independent score matrices in the two lists.

Proof Every normalized layer takes convex combinations of its input token values. By Lemma 1, each network remains in the two-class family, and its coefficient intervals are nested under further layers. Consequently, a scalar coefficient z at any output position after any number of unprimed layers satisfies

$$|z - q_\varepsilon| \le |p_\varepsilon - q_\varepsilon|, \tag{49}$$

because its token stays between $x + p_\varepsilon u$ and $x + q_\varepsilon u$. At the corresponding position in the primed network, the coefficient z' satisfies

$$|z' - q'_\varepsilon| \le |p'_\varepsilon - q'_\varepsilon|, \tag{50}$$

by containment in that network's own token segment, independently of the score matrices used in either list. Both bounds tend to zero, and their reference coefficients q_ε and q'_ε also tend to zero, proving convergence of every output coefficient.

The triangle inequality gives

$$\big|(z - z') - (q_\varepsilon - q'_\varepsilon)\big| \le |p_\varepsilon - q_\varepsilon| + |p'_\varepsilon - q'_\varepsilon|, \tag{51}$$

whose right-hand side is $o(|\delta_\varepsilon|)$ by the two internal-gap hypotheses. Since the repeated coefficients differ by $w\delta_\varepsilon + o(|\delta_\varepsilon|)$, every corresponding pair of output coefficients retains the coefficient w at scale δ_ε, proving Equation (48). Any two unprimed output coefficients differ by at most $|p_\varepsilon - q_\varepsilon|$, while any two primed output coefficients differ by at most $|p'_\varepsilon - q'_\varepsilon|$, so both internal gaps remain $o(|\delta_\varepsilon|)$. These interval bounds apply after every finite number of layers and establish the asserted persistence without a relation between the two lists' score matrices. ■

For the layers with matrices N and N' in Lemma 2, retain y_1, y_2 and y'_1, y'_2 for their output coefficients, and let z, z' be coefficients at corresponding token positions after the remaining layers. Applying the interval bounds to the two internal gaps, each $O(|s_\varepsilon|^3)$, gives

$$z - z' = y_2 - y'_2 + O(|s_\varepsilon|^3), \tag{52}$$

because the remaining normalized layers keep the source tokens between $x+y_1u$ and $x+y_2u$, and keep the competing tokens between $x+y'_1u$ and $x+y'_2u$. For centered probes, $|s_\varepsilon| \le 2|\varepsilon|$ absorbs this change into the $O(|\varepsilon|s_\varepsilon^2)$ remainder in Equation (47), preserving its quadratic coefficient through the remaining layers. A nonzero quadratic coefficient $\kappa_n x^\top (N - N')u$ therefore cannot be canceled by either network's smaller internal gap, even when all remaining source and competing score matrices differ.

B.4. Polynomial nonvanishing

Lemma 4 (Polynomial nonvanishing) *Let $m, d, d_{\text{out}} \ge 1$ and $k \ge 0$ be integers. For every nonzero real polynomial P on $\mathbb{R}^m$, the set*

$$\{z \in \mathbb{R}^m : P(z) \ne 0\} \tag{53}$$

is nonempty, Euclidean open, and dense. In particular, if $L \in \mathbb{R}^{d_{\text{out}} \times d}$ is nonzero and $M_1, \dots, M_k \in \mathbb{R}^{d \times d}$ satisfy $M_j^\top \ne -M_j$ for every $1 \le j \le k$, then

$$P(u) = \|Lu\|^2 \prod_{j=1}^{k} u^\top M_j u \tag{54}$$

is a nonzero polynomial whose nonvanishing set consists precisely of the directions satisfying $Lu \ne 0$ and $u^\top M_j u \ne 0$ for every $1 \le j \le k$. For $k = 0$, the product is one, so the conclusion concerns only the condition $Lu \ne 0$.

Proof Continuity of P makes its nonvanishing set open; density follows once its zero set is shown to have empty interior. We prove the latter assertion by induction on m. A nonzero univariate polynomial has finitely many roots and therefore cannot vanish on an open interval. For $m > 1$, write $z = (v, s) \in \mathbb{R}^{m-1} \times \mathbb{R}$ and expand in the final coordinate as

$$P(v, s) = \sum_{j=0}^{r} a_j(v) s^j, \tag{55}$$

where the coefficient polynomials a_j include at least one nonzero polynomial because P is nonzero. If P vanished on a nonempty open set, it would vanish on a product $U \times I$ with U nonempty and open in $\mathbb{R}^{m-1}$ and I a nonempty open interval. For each $v \in U$, the polynomial $s \mapsto P(v, s)$ would then vanish identically, forcing $a_j(v) = 0$ for every j. A nonzero coefficient polynomial would therefore vanish on U, contradicting the induction hypothesis. Thus the zero set has empty interior, and its complement is dense and nonempty.

For the specialization in Equation (54), the factor $\|Lu\|^2$ is a nonzero polynomial because $L \neq 0$. Each quadratic factor satisfies

$$u^\top M_j u = u^\top \left(\frac{M_j + M_j^\top}{2} \right) u, \tag{56}$$

so it depends only on the symmetric part of M_j. If this quadratic form vanished for every u, evaluation at each coordinate vector would make every diagonal entry of the symmetric part zero, while evaluation at sums of two coordinate vectors would make every off-diagonal entry zero. Consequently $M_j + M_j^\top = 0$, contrary to the hypothesis on M_j. All factors of P are therefore nonzero polynomials, and the absence of zero divisors in the real polynomial ring makes their product nonzero. Since $\|Lu\|^2$ vanishes exactly when $Lu = 0$, nonvanishing of P is equivalent to simultaneous nonvanishing of Lu and every $u^\top M_j u$. Applying the first part of the lemma proves the asserted openness and density, including $k = 0$, when only $\|Lu\|^2$ remains. ■

For recovery of M_r in Theorem 2, apply Lemma 4 to L and the prefix $M_1, \ldots, M_{r-1}$. The condition $Lu \neq 0$ preserves nonzero output displacements along u, while $u^\top M_j u \neq 0$ for $j < r$ prevents collapse through the common prefix. On this dense set of directions, Equation (17) gives $x^\top (M_r - M_r')u = 0$ for every $x \in \mathbb{R}^d$. For each fixed x, continuity in u extends the identity to every direction, and independent variation of x and u determines every entry of $M_r - M_r'$, including its skew-symmetric component.

B.5. Boundary cases

If $t = 1$, row normalization gives a single weight equal to one, so each attention layer fixes its input token independently of the score matrix. If $L = 0$, every score list realizes the zero function. Thus $t \geq 2$ and $L \neq 0$ are necessary for the complete score recovery asserted in Theorem 2.

At two tokens, a skew-symmetric score matrix produces an additional obstruction. For $M^\top = -M$ and input (x, y), write $a = x^\top M y$; the self-scores vanish and the reversed

cross-score is $-a$. Both output tokens coincide because the second satisfies

$$\frac{e^{-a}x + y}{e^{-a} + 1} = \frac{x + e^{a}y}{1 + e^{a}}, \tag{57}$$

whose right-hand side is the first output token, so the output sequence is constant. Every subsequent normalized layer fixes this constant sequence, regardless of its score matrix. In particular, taking

$$M_1 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \tag{58}$$

and the final map $L = I_2$ gives distinct depth-two parameters $W = (M_1, I_2, I_2)$ and $W' = (M_1, 2I_2, I_2)$ with equal complete realizations. All score matrices in this example have full rank, so full rank does not prevent the collapse.

The proper-prefix non-skew condition in Theorem 2 ensures noncollapse on a dense set of probe directions; the two-token example does not establish a converse classification of identifiable parameters. Recovery of M_r requires noncollapse only through $M_1, \ldots, M_{r-1}$, so the terminal matrix M_ℓ is unrestricted. Likewise, $L \neq 0$ supplies a dense set of directions satisfying $Lu \neq 0$ without requiring full rank. No nondegeneracy condition is imposed on unrecovered competing score matrices, because the suffix bounds in Lemma 3 use only convexity.

Appendix C. Formal verification and computational provenance

C.1. AI-assisted development

Codex assists Lean proof search, theorem proving, debugging, inspection of formal code, repository organization, and manuscript drafting. The disclosure in Section 4 records the human and tool contributions. James Amarel is the formalizer and maintainer.

C.2. Statements and correspondence

DeepLean uses the repository's root Lake project. Supporting mathematics is in `DeepLean/Attention/`, with public imports in `DeepLean.lean`. Its principal version pins are

```
Lean 4.35.0-rc2
Mathlib 4.35.0-rc2
  commit 065356127b1dc0016f66b7283ce0ce2c4055aa55
CPython 3.14.5
```

The root `lean-toolchain` and `lake-manifest.json` fix the Lean and library revisions; `scripts/check-local-proof.py` selects the Python version and records the checker identities used for local replay.

One `Challenge.lean`/`Solution.lean` pair represents both numbered theorems through three declarations selected by `comparator.json`. Within this pair, the challenge imports only Mathlib and contains explicit mathematical definitions and the three statement holes. Its definitions mirror `DeepLean/Attention/Definitions.lean`, without importing project proof modules. The solution imports the completed library and proves declarations with the names and types specified in the challenge. Separate exports allow Comparator to compare their statements without importing the challenge holes into the solution.

For Theorem 1, the declaration

```
DeepLean.Attention.normalizer_counterexample
```

produces a scalar normalizer, proves its positivity, normalization at zero, injectivity, Borel measurability, and discontinuity at zero, establishes the nonempty Euclidean-open dense locus $m_1 m_2 l \neq 0$, and supplies at every point the explicit distinct partner $(m_1, 4m_2, l)$ with identical complete realization. Its proof packages the explicit normalizer using the admissibility, discontinuity, locus, and partner lemmas. The supporting endpoint

```
Neuroalgebraic.Submission.Theorem1.theorem1_counterexample
```

states the existential counterexample. Discontinuity at zero for the explicit witness is also included in the selected declaration.

For the pointwise clause of Theorem 2, the declaration

```
DeepLean.Attention.softmax_identifiability
```

requires a nonzero source final map, a non-skew proper source score prefix, equal positive depths, and equality of the complete realization functions at positive temperature with $t \geq 2$. No nondegeneracy condition is imposed on the terminal source score or the competing parameter, and the conclusion identifies the complete virtual parameter points. Natural-number dimensions are allowed formally; the nonzero final map excludes zero-dimensional cases. Its supporting endpoint

```
Neuroalgebraic.Submission.Theorem2.theorem2_softmax_identifiability
```

has the identical pointwise statement.

For the explicit generic clause, the declaration

```
DeepLean.Attention.softmax_generic_identifiability
```

states nonemptiness, Euclidean openness, and density of the coordinate locus in Equation (9), its containment in the source set in Equation (7), and singleton complete fibers throughout it. A fixed-depth parameter with tail length k has depth $k+1$; at depth one the proper-prefix condition is empty. Existence of the coordinate indices witnesses positive input and output dimensions, while the fiber ranges over all parameters of the fixed depth. The generic wrapper combines the genericity and fiber proofs with `NormalizedDeepAttention.properCoordinateRegular_properDeep`, making the manuscript's containment clause explicit. Its supporting endpoint

```
Neuroalgebraic.Submission.Theorem2.theorem2_openDense_coordinate_fibers
```

provides the openness, density, and singleton-fiber assertions.

The architecture reduction is supplied by

```
NormalizedDeepAttention.deepNormalizedAttention_eq_virtual
```

which identifies the primitive and virtual realizations.

Log-affine reparametrization and boundary analysis are supplied by

```
Neuroalgebraic.Submission.DeepAttentionVirtual.
  logAffine_properPrefix_pointwise_identifiable
  logAffine_properCoordinate_fullFiber
  properCoordinateLocus_nonempty_open_dense
  skewBoundary_nontrivialFiber
```

which remain supporting library results; Palomar comparison selects only the three `DeepLean.Attention` declarations corresponding to the numbered theorems.

C.3. Manuscript-facing Lean declarations

Each theorem listing is imported directly from `Solution.lean`, so the manuscript reproduces the source declaration without maintaining a separate copy.

Theorem 1.

```
theorem normalizer_counterexample :
    ∃ S : ℝ → ℝ,
      (∀ x, 0 < S x) ∧
      S 0 = 1 ∧
      Function.Injective S ∧
      Measurable S ∧
      ¬ ContinuousAt S 0 ∧
      Set.Nonempty parameterLocus ∧
      IsOpen parameterLocus ∧
      Dense parameterLocus ∧
      ∀ p ∈ parameterLocus,
        source p ≠ partner p ∧
        realizeDeep (t := 2) S (fixedDepthToDeep (partner p)) =
          realizeDeep S (fixedDepthToDeep (source p)) :=
  ⟨exoticNormalization, normalizer_admissible.1,
    normalizer_admissible.2.1, normalizer_admissible.2.2.1,
    normalizer_admissible.2.2.2, normalizer_discontinuous_at_zero,
    parameterLocus_nonempty_open_dense.1,
    parameterLocus_nonempty_open_dense.2.1,
    parameterLocus_nonempty_open_dense.2.2, explicit_partner⟩
```

Theorem 2: pointwise identifiability.

```
theorem softmax_identifiability {d dOut t : ℕ}
    (τ : ℝ) (hτ : 0 < τ) (ht : 2 ≤ t)
    (W W' : DeepWeights d dOut) (hL : W.L ≠ 0)
    (hscore : ∀ M ∈ (W.M₁ :: W.rest).dropLast, M.transpose ≠ -M)
    (hdepth : W.rest.length = W'.rest.length)
    (hEq : realizeDeep (t := t) (temperatureNormalization τ) W =
      realizeDeep (temperatureNormalization τ) W') : W = W' :=
  Neuroalgebraic.Submission.Theorem2.theorem2_softmax_identifiability
    τ hτ ht W W' hL hscore hdepth hEq
```

Theorem 2: open-dense singleton fibers.

```
theorem softmax_generic_identifiability {d dOut k t : ℕ}
    (τ : ℝ) (hτ : 0 < τ) (ht : 2 ≤ t) (i : Fin d) (o : Fin dOut) :
    Set.Nonempty {W : FixedDepthWeights d dOut k | ProperCoordinateRegular i o W} ∧
      IsOpen {W : FixedDepthWeights d dOut k | ProperCoordinateRegular i o W} ∧
      Dense {W : FixedDepthWeights d dOut k | ProperCoordinateRegular i o W} ∧
      ∀ W : FixedDepthWeights d dOut k, ProperCoordinateRegular i o W →
        ((fixedDepthToDeep W).L ≠ 0 ∧
          ∀ M ∈ ((fixedDepthToDeep W).M₁ :: (fixedDepthToDeep W).rest).dropLast,
            M.transpose ≠ -M) ∧
        FixedDepthFiberFor (t := t) τ W = {W} :=
  ⟨properCoordinateRegular_nonempty i o,
    isOpen_properCoordinateRegular i o,
    dense_properCoordinateRegular i o,
    fun W hW ↦ ⟨properCoordinateRegular_properDeep i o W hW,
      Neuroalgebraic.Submission.Theorem2.theorem2_coordinate_fullFiber
        τ hτ ht i o W hW⟩⟩
```

C.4. Reproduction and evidence boundary

With the pinned Lean toolchain installed through `elan` and the root manifest's dependencies prepared, the complete local check is invoked from the repository root by

```
python3 scripts/check-local-proof.py
```

which creates a fresh evidence directory under `.cache/local-proof/` and records source hashes, tool identities, commands, process exits, and a final `report.json`. The check validates the mirrored definitions and declaration signatures, builds the project, elaborates Challenge and Solution separately, scans completed sources for placeholders, and audits the transitive axiom dependencies of all three selected declarations. `Challenge.lean` specifies the theorem statements with proof placeholders, while `Solution.lean` supplies proofs containing no placeholders, directly or through their dependencies. The permitted axioms are exactly `propext`, `Quot.sound`, and `Classical.choice`.

For proof replay, the script exports the selected declarations and required primitive targets with `leanexport`. Comparator compares the Challenge and Solution statements and checks the axiom policy under `comparator.json`, invoking NanoDa and Lean's checker; `con-ron`, an external Lean proof checker implemented in Rust, separately checks the Solution export in verified mode. Acceptance requires results identifying every selected declaration and every required checker, in addition to successful process exits.

The retained source checksums, tool versions, commands, and results are indexed in `docs/verification/README.md`.